



Try MySQL HeatWave for free

One MySQL Database service for transactions, analytics, and machine learning.

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Databases are the essential data repository for all software applications. For example, whenever someone conducts a web search, logs in to an account, or completes a transaction, a database system is storing the information so it can be accessed in the future.

A **relational database** stores data in separate tables rather than putting all the data in one big storeroom. The database structure is organized into physical files optimized for speed. The logical data model, with objects such as data tables, views, rows, and columns, offers a flexible programming environment. You set up rules governing the relationships between different data fields, such as one to one, one to many, unique, required, or optional, and “pointers” between different tables. The database enforces these rules so that with a well-designed database your application never sees data that’s inconsistent, duplicated, orphaned, out of date, or missing.

The “SQL” part of “MySQL” stands for “Structured Query Language.” SQL is the most common standardized language used to access databases. Depending on your programming environment, you might enter SQL directly (for example, to generate reports), embed SQL statements into code written in another language, or use a language-specific API that hides the SQL syntax.

MySQL is open source

Open source means it’s possible for anyone to use and modify the software. Anybody can download MySQL software from the internet and use it without paying for it. You can also change its source code to suit your needs. MySQL software uses the [GNU General Public License \(GPL\)](#) to define what you may and may not do with the software in different situations.

If you feel uncomfortable with the GNU GPL or need to embed MySQL code into a commercial application, you can buy a commercially licensed version from Oracle. See the [MySQL Licensing Information section](#) for more information.

MySQL: the #1 choice for developers

MySQL consistently ranks as the most popular database for developers, according to [Stack Overflow](#) and [JetBrains](#). Developers love its high performance, reliability, and

MySQL supports the following popular development languages and drivers:

PHP	Python	Java/JDBC	Node.js
Perl	Ruby	Go	Rust
C	C++	C#/.NET	ODBC

MySQL has also become the database of choice for many of the most successful open source applications, including WordPress, Drupal, Joomla, and Magento. MySQL is the “M” in the highly popular open source LAMP (Linux, Apache, MySQL, Perl/Python/PHP) stack to develop web applications.

MySQL works in client/server or embedded systems

MySQL Database is a client/server system that consists of a multithreaded SQL server that supports different back ends, several different client programs and libraries, administrative tools, and a wide range of application-programming interfaces (APIs). We also provide MySQL as an embedded multithreaded library that you can link into your application to get a smaller, faster, easier-to-manage standalone product.

MySQL benefits

MySQL is fast, reliable, scalable, and easy to use. It was originally developed to handle large databases quickly and has been used in highly demanding production environments for many years.

Although MySQL is under constant development, it offers a rich and useful set of functions. MySQL’s connectivity, speed, and security make it highly suited for accessing databases on the internet.

MySQL’s key benefits include

Ease of use: Developers can install MySQL in minutes, and the database is easy to manage.

Reliability: MySQL is one of the most mature and widely used databases. It has been tested in a wide variety of scenarios for more than 25 years, including by many of the world’s largest companies. Organizations depend on MySQL to run business-critical applications because of its reliability.

Scalability: MySQL scales to meet the demands of the most accessed applications. native replication architecture enables organizations such as Facebook to scale applications to support billions of users.

Performance: MySQL HeatWave is **faster and less expensive** than other database systems, as demonstrated by multiple standard industry benchmarks, including TPC-H, TPC-DS, and TPC-Benchmark.

High availability: MySQL delivers a complete set of native, fully integrated replication technologies for high availability and disaster recovery. For business-critical applications, and to meet service-level agreement commitments, customers can achieve

Recovery point objective = 0 (zero data loss)

Recovery time objective = seconds (automatic failover)

Security: [Data security](#) entails protection and compliance with industry and government regulations, including the European Union General Data Protection Regulation, the Payment Card Industry Data Security Standard, the Health Insurance Portability and Accountability Act, and the Defense Information Systems Agency's Security Technical Implementation Guides. MySQL Enterprise Edition provides advanced security features, including authentication/authorization, transparent data encryption, auditing, data masking, and a database firewall.

Flexibility: The MySQL Document Store gives users maximum flexibility in developing traditional SQL and NoSQL schema-free database applications. Developers can mix and match relational data and JSON documents in the same database and application.

MySQL use cases

Cloud applications: MySQL is very popular in the cloud. [MySQL HeatWave](#) is a fully managed database service, powered by the integrated HeatWave in-memory query accelerator. It's the only cloud database service that combines transactions, real-time analytics across data warehouses and data lakes, and machine learning (ML) services into one MySQL Database—without the complexity, latency, cost, and risk of ETL duplication. MySQL HeatWave is 6.5X faster than Amazon Redshift at half the cost, 7X faster than Snowflake at one-fifth the cost, and 1,400X faster than Amazon Aurora at half the cost. With MySQL HeatWave AutoML, developers and data analysts can build, train, deploy, and explain [machine learning](#) models within MySQL HeatWave in a fully automated way—25X faster than Amazon Redshift ML at 1% of the cost.



MySQL is extremely popular for

Ecommerce: Many of the world's largest ecommerce applications (for example, Shopify, Uber, and Booking.com) run their transactional systems on MySQL. It's a popular choice for managing user profiles, credentials, user content, financial data including payments, and fraud detection.

Social platforms: Facebook, Twitter, and LinkedIn are among the world's largest social networks that rely on MySQL.

Content management: Unlike single-purpose document databases, MySQL enables both SQL and NoSQL with a single database. The MySQL Document Store enables CRUD operations and the power of SQL to query data from JSON documents for reporting and analytics.

SaaS and ISVs: More than 2,000 ISVs, OEMs, and VARs, including Ericsson, F5, and IBM, rely on MySQL as their embedded database to make their applications, hardware, and appliances more competitive, bring them to market faster, and lower their cost of goods sold. MySQL is also the database behind popular SaaS applications, including Zendesk and HubSpot.

Other popular applications using MySQL include ones for online gaming, digital marketing, retail point-of-sale systems, and Internet of Things monitoring systems.

On-premises applications with MySQL Enterprise Edition: MySQL Enterprise Edition includes the most comprehensive set of advanced features, management tools, and technical support to achieve the highest levels of MySQL scalability, security, reliability, and uptime. It reduces the risk, cost, and complexity in developing, deploying, and managing business-critical MySQL applications. It provides security features, including MySQL Enterprise Backup, Monitor, Firewall, Audit, Transparent Data Encryption, and Authentication, to help customers protect data and achieve regulatory and industry compliance.

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